Instruction to the candidate:

Write your name and IC number in the space provided.

1. Candidates shall only start reading the questions when they are allowed to do so.
2. Answer all questions.
3. No notes were allowed in the examination hall unless informed to do so.
4. Cheating is prohibited.
5. You cannot bring this question paper out from this hall.
6. You shall not use the Internet unless you are about to submit your work.
7. Candidates must complete the task given in the time provided.

THIS PAPER CONSISTS OF 22 PRINTED PAGES.

SECTION A: TITLE: BUILD A REACT STREET FOOD MENU APP WITH SEARCH, FILTERS, SORTING AND PAGINATION

A. TASK: Your primary task is to build a single-page React App for Makan Trail, a Malaysian street food guide. The app shows hawker dishes, desserts and drinks in one table that the user can search, filter, sort and paginate, based on the data received from the provided back-end web server.

B. INSTRUCTIONS:

1. Begin by downloading the project files from the link provided by your assessor. There will be two zipped folders containing the front-end and the back-end project. After downloading these files, unzip them.
2. Project Structure:
    a. **b18-retake-app-qv**: This folder contains the React app you will work on.
    b. **b18-retake-api**: This folder contains the Express.js back-end. **DO NOT MODIFY** the contents of this folder.
3. Setup and start the development environments:
    a. Open both folders in your Visual Studio Code.
    b. Run the npm install command in each folder to install all required packages.
    c. Start the back-end server by running the npm start command in the b18-retake-api folder. The API will run on http://localhost:5123.
    d. Then, start the front-end development server by running the npm run dev command in the b18-retake-app-qv folder. The app will run on http://localhost:5173.
4. Development Instructions:
    a. Focus your development within the b18-retake-app-qv folder.
    b. Look for the sections in the code marked with the comment **"INSTRUCTION:"** in the b18-retake-app-qv folder. Complete the tasks specified at each of these markers.
    c. The back-end API will handle all filtering, sorting, searching and pagination. Your front-end should be designed to send appropriate

requests to the API and handle the responses to update the UI dynamically. **Do not** fetch all the records and then filter or sort them inside the browser.

d. Implement the following features:
    i. **Menu Table**: Display every menu item — hawker dishes, desserts and drinks — in one table, showing the name, course, state, main ingredient, calories and price.
    ii. **Search and Filters**: Add a search box (by name or main ingredient) and dropdowns to filter by state (Penang, Kuala Lumpur, Melaka, Perak, Kelantan, Sarawak) and by course (Main, Dessert, Drink). Both filters can be applied together.
    iii. **Sorting and Pagination**: Sort the table by name or price, choose how many items are shown per page, and move between pages.
    iv. **Loading States:** Show loading indicators while fetching data.
    v. **Error Handling:** Display appropriate error messages when API calls fail, and a suitable message when no records match.

5. Submission Requirements
    a. Submit the completed b18-retake-app-qv folder according to the instructions given by your assessor. Do not include the node_modules folder in your submission.

API Documentation

The provided back-end runs on http://localhost:5123 and exposes one main endpoint, plus a helper endpoint. The main endpoint supports search, filtering, sorting and pagination, and they can all be combined in one request. The same documentation is available in the README.md file inside the b18-retake-api folder.

Features

- **Search**: by name or main ingredient (e.g., “noodle”, “coconut”)

- **Filter**: by state (e.g., “Penang”, “Sarawak”) and by course (“Main”, “Dessert” or “Drink”). Both filters are combinable.

- **Sort and paginate**: sort by name or price, and paginate results (e.g., 5, 10, 20 per page)

API Endpoints

1. Menu Endpoint

- Route: `GET /menu`
- Description: Returns the street food menu: hawker dishes, desserts and drinks from six Malaysian states. Supports search, filtering, sorting and pagination.
- Query Parameters:
- `search` (optional): matches either the `name` or the `mainIngredient` (case-insensitive, partial match).
- `state` (optional): Must be one of `“Penang”`, `“Kuala Lumpur”`, `“Melaka”`, `“Perak”`, `“Kelantan”`, or `“Sarawak”`.
- `course` (optional): Must be one of `“Main”`, `“Dessert”`, or `“Drink”`.
- `sort` (optional): `“name”` or `“price”`.
- `page` (optional): integer, default is `1`.
- `limit` (optional): integer, default is `10`.

Searching and Filtering

- `search`, `state` and `course` can be used on their own or together; each one narrows the results further.
- If you omit all three, all 62 items are returned (one page at a time).

Sorting

- If `sort=name`, results are sorted alphabetically by name (ascending, A to Z).
- If `sort=price`, results are sorted by price from the cheapest to the most expensive. Items with the same price are sorted by name.

Pagination

- There are 62 items in total. `total` and `totalPages` in the response count only the items that match the search and filters.
- Default: 10 items per page (`page=1`, `limit=10`).
- You can customize these parameters, e.g., `page=2`, `limit=5`.

Example:

GET /menu?
search=coconut&course=Dessert&sort=price&page=1&limit=5

This returns the first page (5 per page) of desserts whose name or main ingredient contains “coconut”, sorted from the cheapest. The response below is shortened to the first item on that page.

Important: this endpoint does not return a bare array. The items are inside the `data` property, `total` is the number of matching items, and `totalPages` tells you how many pages there are.

Response Example:

{
"page": 1, "limit": 5, "total": 6, "totalPages": 2,
"data": [
{ "id": 36, "name": "Puteri Mandi", "course": "Dessert", "state": "Kelantan",
"mainIngredient": "Glutinous rice and grated coconut", "calories": 190,
"price": 2 }
]
}

2. States Endpoint (helper)

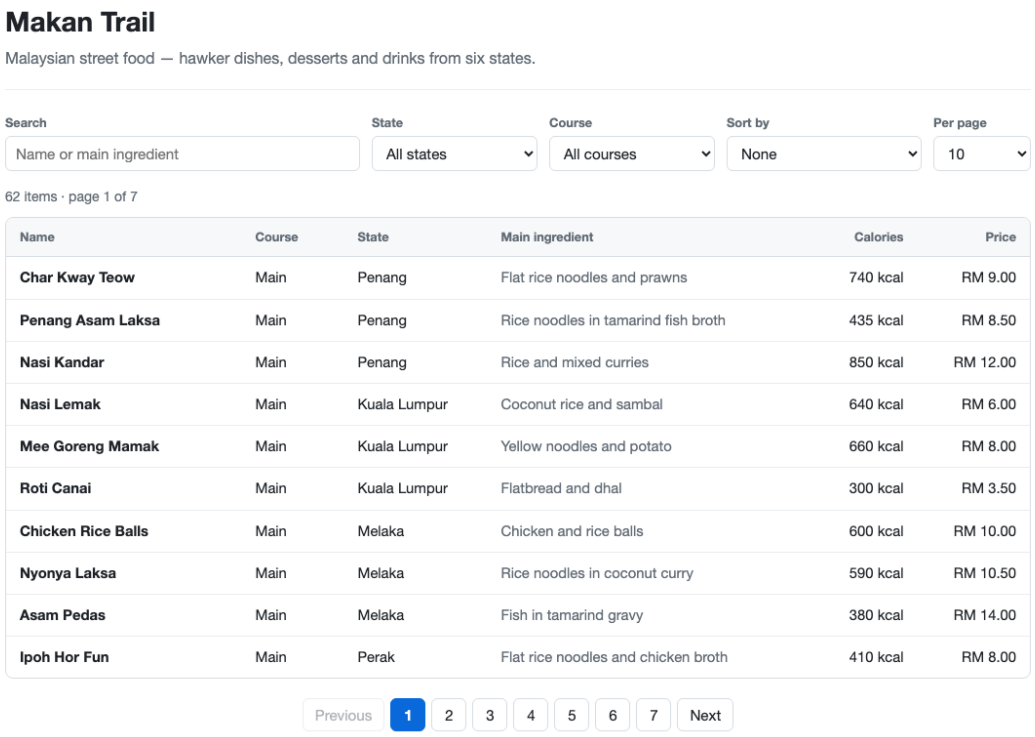
- Route: `GET /states`
- Description: Returns the list of states used in the menu. You may use this to build your state dropdown instead of hard-coding the options.

Response Example:

["Kelantan", "Kuala Lumpur", "Melaka", "Penang", "Perak", "Sarawak"]

C. DRAWING, DATA, AND SCHEDULE:

The figures below show the expected functionality of the application. They are a guide to the features and the information that must be displayed — your own styling and colour scheme do not have to match them exactly.



Makan Trail

Malaysian street food — hawker dishes, desserts and drinks from six states.

Search: Name or main ingredient | State: All states | Course: All courses | Sort by: None | Per page: 10

62 items · page 1 of 7

Name	Course	State	Main ingredient	Calories	Price
Char Kway Teow	Main	Penang	Flat rice noodles and prawns	740 kcal	RM 9.00
Penang Asam Laksa	Main	Penang	Rice noodles in tamarind fish broth	435 kcal	RM 8.50
Nasi Kandar	Main	Penang	Rice and mixed curries	850 kcal	RM 12.00
Nasi Lemak	Main	Kuala Lumpur	Coconut rice and sambal	640 kcal	RM 6.00
Mee Goreng Mamak	Main	Kuala Lumpur	Yellow noodles and potato	660 kcal	RM 8.00
Roti Canai	Main	Kuala Lumpur	Flatbread and dhal	300 kcal	RM 3.50
Chicken Rice Balls	Main	Melaka	Chicken and rice balls	600 kcal	RM 10.00
Nyonya Laksa	Main	Melaka	Rice noodles in coconut curry	590 kcal	RM 10.50
Asam Pedas	Main	Melaka	Fish in tamarind gravy	380 kcal	RM 14.00
Ipoh Hor Fun	Main	Perak	Flat rice noodles and chicken broth	410 kcal	RM 8.00

Previous 1 2 3 4 5 6 7 Next

Figure 1: Menu page (default view — no search, filter or sorting, 10 items per page)

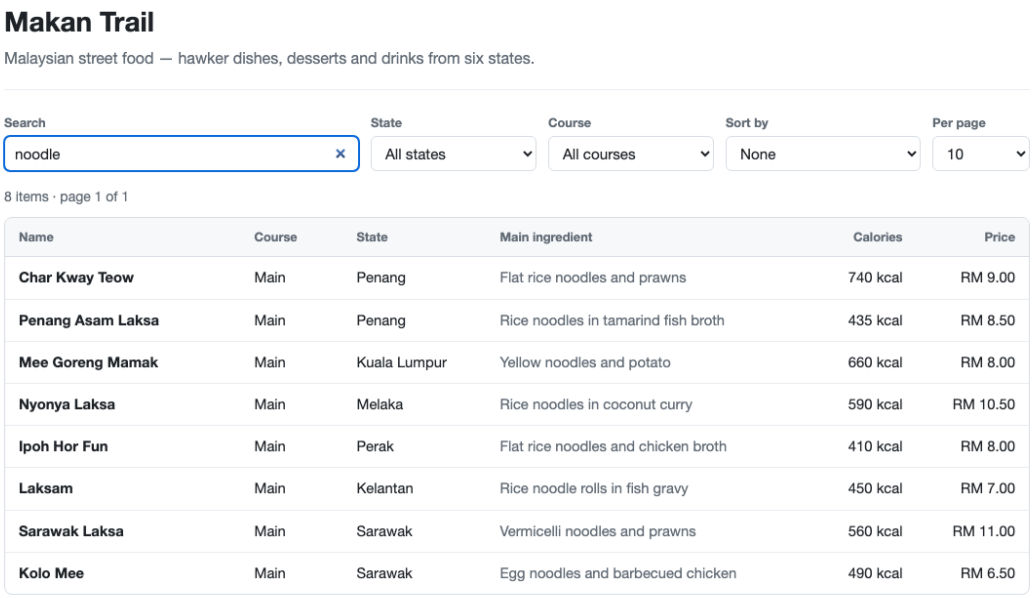


Figure 2: Menu page (search “noodle”)

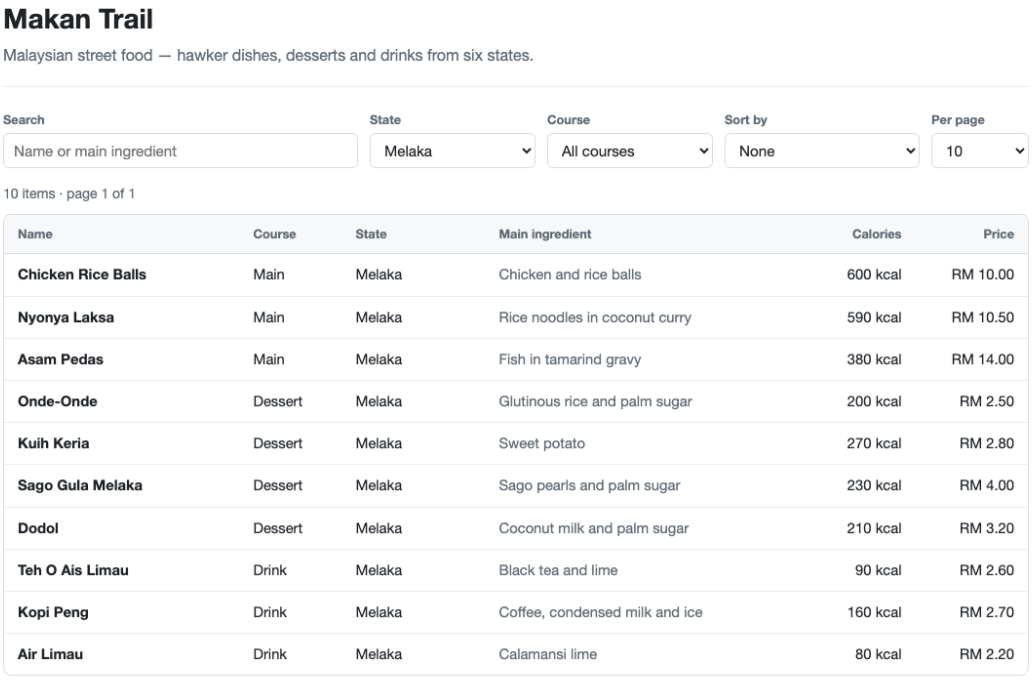


Figure 3: Menu page (filtered by “Melaka” state)

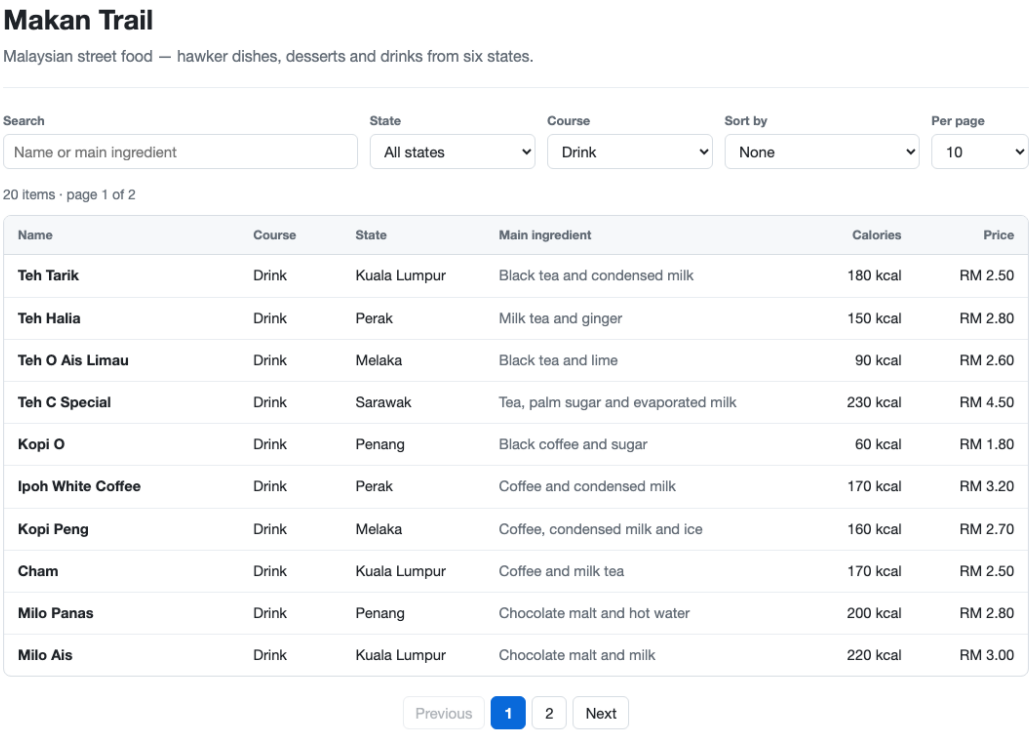


Figure 4: Menu page (filtered by “Drink” course)

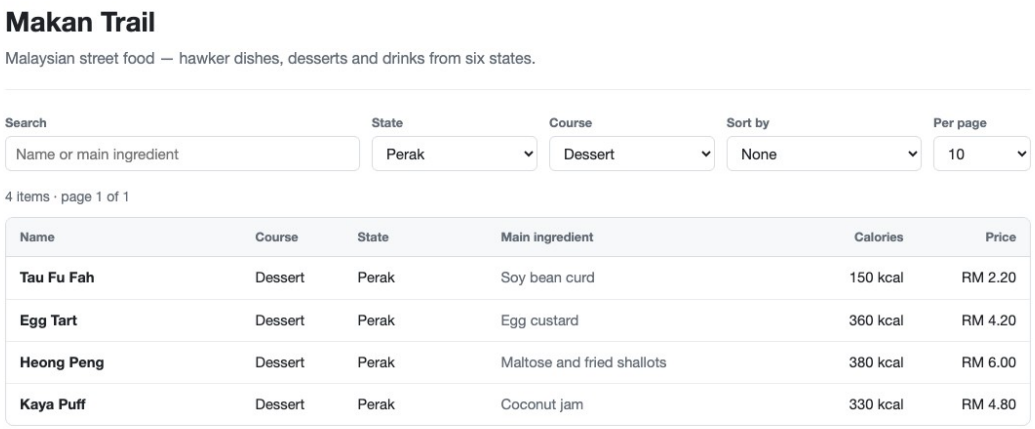


Figure 5: Menu page (filtered by “Perak” state and “Dessert” course — both filters applied together)

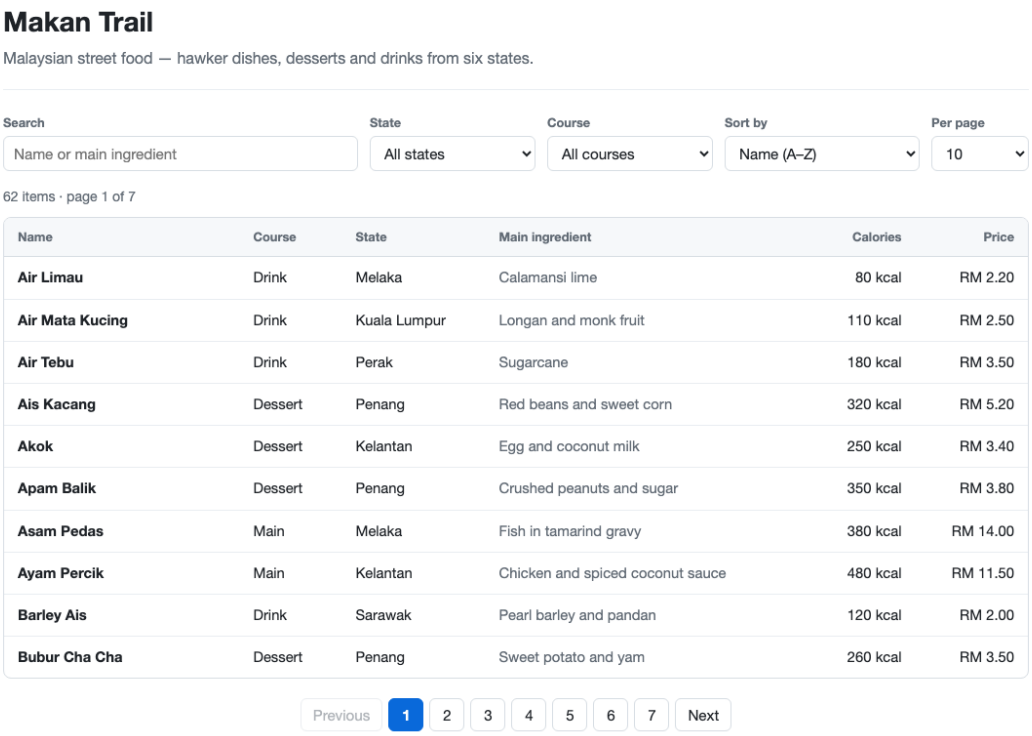


Figure 6: Menu page (sorted by name — A to Z)

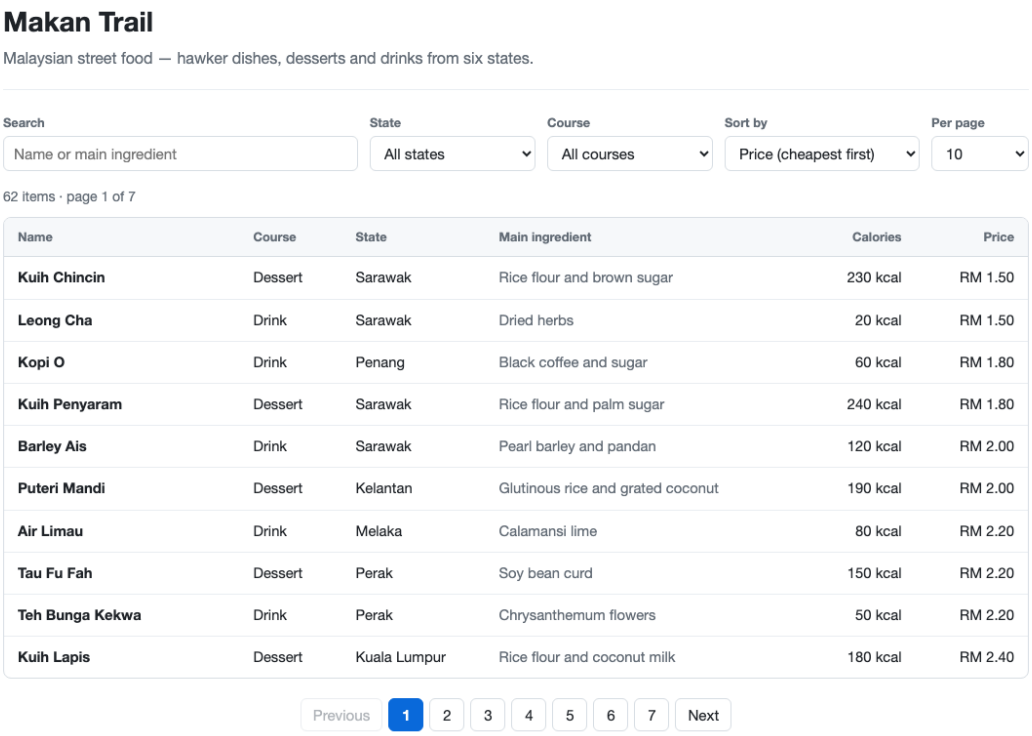
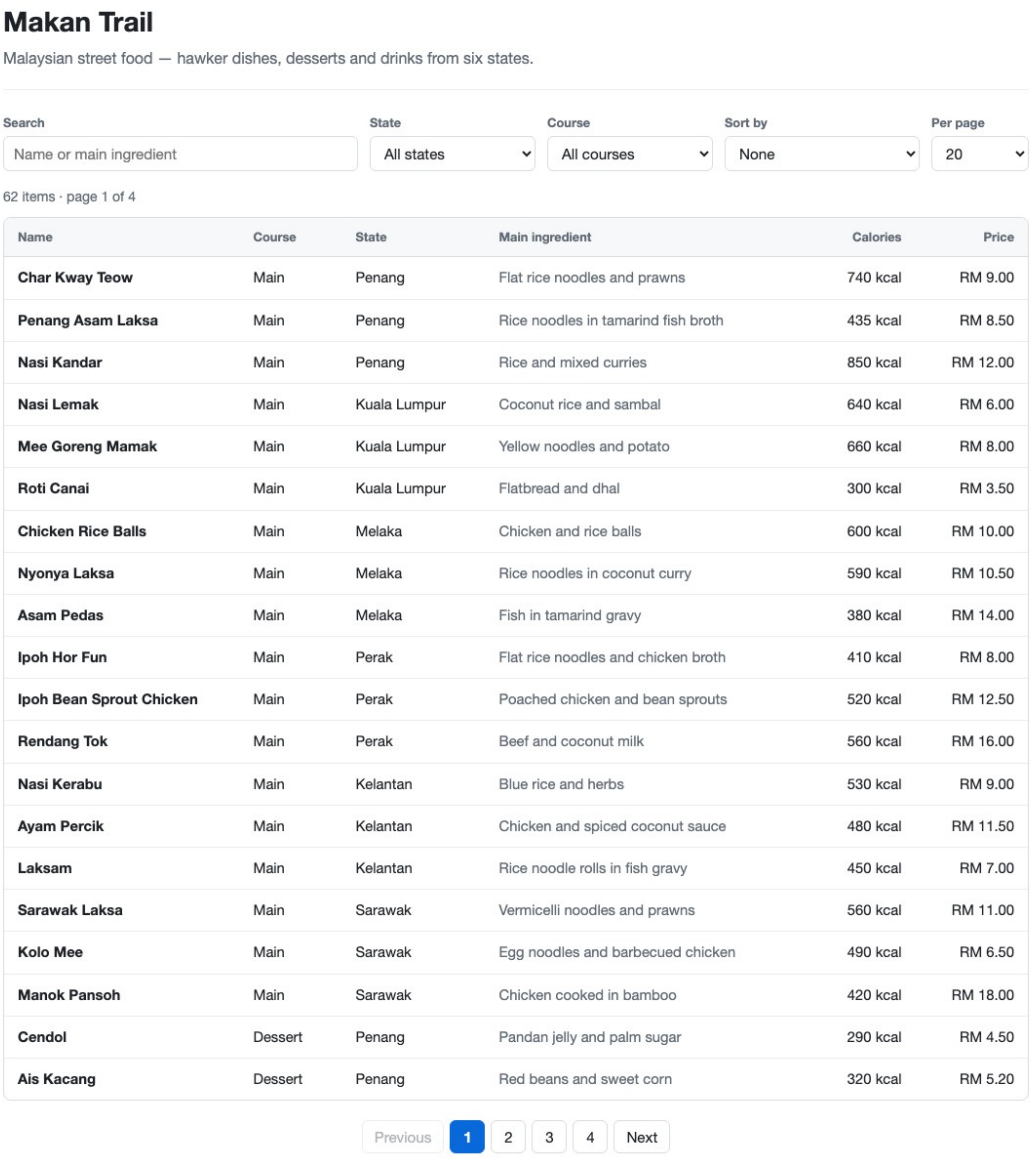


Figure 7: Menu page (sorted by price — cheapest first)



Makan Trail

Malaysian street food — hawker dishes, desserts and drinks from six states.

Search: Name or main ingredient | State: All states | Course: All courses | Sort by: None | Per page: 20

62 items · page 1 of 4

Name	Course	State	Main ingredient	Calories	Price
Char Kway Teow	Main	Penang	Flat rice noodles and prawns	740 kcal	RM 9.00
Penang Asam Laksa	Main	Penang	Rice noodles in tamarind fish broth	435 kcal	RM 8.50
Nasi Kandar	Main	Penang	Rice and mixed curries	850 kcal	RM 12.00
Nasi Lemak	Main	Kuala Lumpur	Coconut rice and sambal	640 kcal	RM 6.00
Mee Goreng Mamak	Main	Kuala Lumpur	Yellow noodles and potato	660 kcal	RM 8.00
Roti Canai	Main	Kuala Lumpur	Flatbread and dhal	300 kcal	RM 3.50
Chicken Rice Balls	Main	Melaka	Chicken and rice balls	600 kcal	RM 10.00
Nyonya Laksa	Main	Melaka	Rice noodles in coconut curry	590 kcal	RM 10.50
Asam Pedas	Main	Melaka	Fish in tamarind gravy	380 kcal	RM 14.00
Ipoh Hor Fun	Main	Perak	Flat rice noodles and chicken broth	410 kcal	RM 8.00
Ipoh Bean Sprout Chicken	Main	Perak	Poached chicken and bean sprouts	520 kcal	RM 12.50
Rendang Tok	Main	Perak	Beef and coconut milk	560 kcal	RM 16.00
Nasi Kerabu	Main	Kelantan	Blue rice and herbs	530 kcal	RM 9.00
Ayam Percik	Main	Kelantan	Chicken and spiced coconut sauce	480 kcal	RM 11.50
Laksam	Main	Kelantan	Rice noodle rolls in fish gravy	450 kcal	RM 7.00
Sarawak Laksa	Main	Sarawak	Vermicelli noodles and prawns	560 kcal	RM 11.00
Kolo Mee	Main	Sarawak	Egg noodles and barbecued chicken	490 kcal	RM 6.50
Manok Pansoh	Main	Sarawak	Chicken cooked in bamboo	420 kcal	RM 18.00
Cendol	Dessert	Penang	Pandan jelly and palm sugar	290 kcal	RM 4.50
Ais Kacang	Dessert	Penang	Red beans and sweet corn	320 kcal	RM 5.20

Previous | 1 | 2 | 3 | 4 | Next

Figure 8: Menu page (with pagination, limited to 20 items per page)

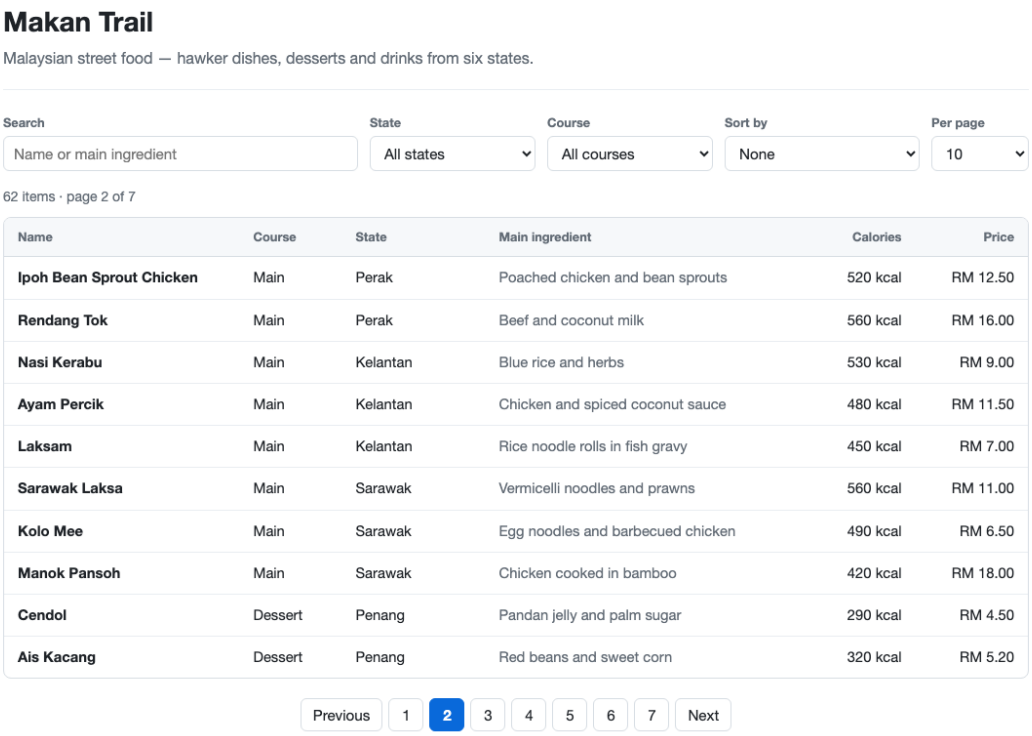


Figure 9: Menu page (page 2 selected)

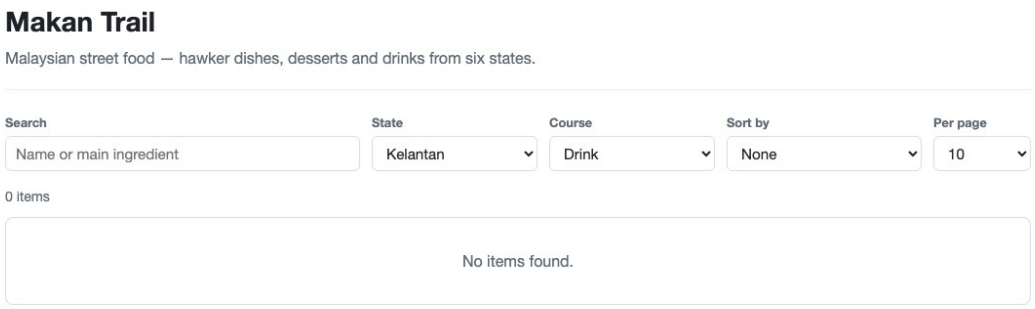


Figure 10: Menu page (no matching items — filtered by “Kelantan” state and “Drink” course)

D. TOOLS, EQUIPMENT DAN MATERIALS

NO.	MATERIALS	QUANTITY (MATERIALS:CANDIDATE)
-	-	-
NO.	**EQUIPMENT**	**QUANTITY (EQUIPMENT:CANDIDATE)**
1	Laptop	1:1
2	Table	1:1
3	Chair	1:1
4	Text Editor (VS Code)	1:1
5	NodeJS	1:1

E. DURATION: 3 HOURS

F. ASSESSMENT CRITERIA

The candidate's performance will be assessed according to the following criteria:

1. **Critical task:**
    a. **Display List:** Render a table that shows the menu items fetched from the back-end. At minimum, each row should display (name, course, state, mainIngredient, calories, price).
    b. **Dynamic Filtering:** Implement dropdowns that allow users to filter the menu by state (Penang, Kuala Lumpur, Melaka, Perak, Kelantan, Sarawak) and by course (Main, Dessert, Drink), with the results fetched dynamically from the back-end. Both filters must work together.
    c. **Sort Functionality:** Add options above the table to sort the menu by name or price using a dropdown menu. The sorting requests should be sent to and processed by the back-end. By default, there won’t be any sorting. The dropdown menu should show “none” as the default option.
    d. **Search Capability:** Include a search box above the table that allows users to find menu items by their name or their main ingredient. The search query should be processed by the back-end to return relevant results.
    e. **Pagination:** Below the table, implement pagination controls that interact with the back-end to fetch a specific page of data. Include options to set

how many items are displayed per page (5, 10 and 20).

2. **Task Process:**
    a. **Component and Layout Structure:** Organise the app into clear React components, building on the provided structure (the header in App, the menu in Menu and the API call in utils/api.js), and lay out the page as shown in the figures: the controls above the table and the pagination below it.
    b. **Interactive Functionality:** Make the search box, both filters, the sort and the page-size options work together, so that changing one keeps the others. Changing any of them should return the table to page 1, and the current page should be highlighted.
    c. **User Interface Design:** Develop a user interface that is visually appealing, intuitive, and user-friendly, demonstrating good UI/UX principles.

3. **Task Outcome:**
    a. **Code Quality:** The code should be clean, well-commented, efficient, and adhere to modern best practices in React development.
    b. **Loading, Error and Empty States:** The application should show a loading indicator while data is being fetched, an appropriate error message when the API cannot be reached, and a suitable message when no records match the current filters.

4. **Attitude/Safety/Environment**
    a. Submit the solution according to the guidelines.
    b. Attend the assessment on time.

Performance Assessment Scheme:

Name of Candidate:	**No. IC:**
Start Time: **End Time:**	**Date:**

Marking scale:

0=Underperform/Wrong; 1=Unsatisfactory ; 2=Satisfactory ; 3=Above Expectation

Marking Calculation:

	% **The weighting will be determined by the expert who developed the question**

Section	**ASSESSMENT CRITERIA**	**MARKS**	
1.	**Critical Task (50%)**	**Mark a (/) on the acceptable and non-acceptable on the following critical task.**	
		Acceptable	Non-acceptable
	1.1 Completed the table with all the columns for the menu items (name, course, state, main ingredient, calories and price), fetched from the back-end.		
	1.2 Completed the “Filter by State” and “Filter by Course” functionality, with both filters fetched and processed by the back-end.		
	1.3 Completed the sort functionality: a dropdown with “none” as the default option that sorts the menu by name (A to Z) or by price (cheapest first), with the sorting processed by the back-end.		

	1.4 Completed the search functionality: a search box that finds menu items by their name or main ingredient, with the search processed by the back-end.			
	1.5 Completed the pagination functionality: controls to move between pages and a choice of 5, 10 or 20 items per page, with each page fetched from the back-end (page, limit and totalPages).			
	(*Critical criteria: 50 marks will be given if the above criteria task was performed. 0 marks will be given if any of the criteria tasks were not carried out.)			
	Marks obtained (%)		**50**	
2.	**Task Process (30%)**		**% Weightage**	**MARKS**
	2.1 Component and Layout Structure			
	i. Well structured beyond the provided project: the header is in App, the fetch is in utils/api.js, parts such as the pagination or the controls are their own components, and the layout matches the figures (controls above the table, pagination below).	3	**10%**	
	ii. Follows the provided project's structure (header in App, table and controls in Menu, fetch in utils/api.js) and the layout in the figures, with minor issues at most.	2		
	iii. The page works, but the structure is weak: e.g., the fetch is written inside the component, there is no header, or the controls and pagination are not where the figures show them.	1		
	iv. The table and controls are not rendered as a working page.	0		
	2.2 Interactive Functionality			
	i. Search, both filters, sort and page size all work together; changing one keeps the others and returns to page 1, and the current page is highlighted.	3	**10%**	
	ii. The controls work together, with minor issues (e.g., the page does not return to 1, or the current page is not highlighted).	2		

	iii. Each control works on its own, but they do not work together (e.g., choosing a filter clears the search or the sort).	1		
	iv. The controls do not respond.	0		
	2.3 User Interface Design			
	i. Excellent user interface design, featuring exceptional visual appeal, intuitiveness, and a high degree of user-friendliness.	3	**10%**	
	ii. Good user interface design, displaying visual appeal, intuitiveness, and a satisfactory level of user-friendliness.	2		
	iii. Basic user interface design with some visual appeal and intuitiveness, but improvements are needed in terms of overall user-friendliness.	1		
	iv. Poor user interface design, lacking visual appeal, intuitiveness, and overall user-friendliness.	0		
	Marks obtained (%)			
3.	**Task Outcome (15%)**			
	3.1 Code Quality			
	i. The code is exemplary: readable, efficient, and adheres to best practices	3	**10%**	
	ii. The code is mostly good but has minor readability or efficiency issues	2		
	iii. The code has significant readability or efficiency issues	1		
	iv. The code is unreadable or inefficient, failing to meet basic coding standards	0		
	3.2 Loading, Error and Empty States			
	i. Loading indicators, API error messages and no-result messages are all present and behave correctly.	3	**5%**	
	ii. Most of these states are handled, but one is missing or behaves inconsistently.	2		
	iii. Only a loading indicator is present; API errors or empty results are not handled.	1		
	iv. No loading, error or empty-state handling at all.	0		

	Marks obtained (%)			
4.	**Attitude/ Safety/ Environment (5%)**			
	4.1 Submit the solution according to the guideline			
	i. The solution is submitted in full accordance with the guidelines, demonstrating exceptional attention to detail and adherence to assessment requirements	3	**3%**	
	ii. The solution is submitted mostly in accordance with the guidelines, demonstrating satisfactory compliance with assessment requirements and minor deviations	2		
	iii. The solution is submitted with significant deviations from the guidelines, demonstrating an unsatisfactory understanding of the assessment requirements	1		
	iv. The solution is submitted without regard for the guidelines or not submitted at all, demonstrating a lack of understanding or commitment to the assessment requirements	0		
	4.2 Attend the Assessment on Time			
	i. Arrives well ahead of the scheduled assessment time, demonstrating exceptional punctuality and time management skills	3	**2%**	
	ii. Arrives slightly late (within 10 min) for the scheduled assessment, demonstrating satisfactory punctuality and time management skills.	2		
	iii. Arrives slightly late (more than 15 min) for the scheduled assessment, demonstrating unsatisfactory punctuality and time management skills.	1		
	iv. Arrives significantly late (more than 30min) or does not attend the scheduled assessment, demonstrating a lack of punctuality and time management skills.	0		
	Marks obtained (%)			

Total Marks Combine **(Section 1 + Section 2 + Section 3 + Section 4)**	
Assessment Result	
MARKS:	**SHOWN/ NOT SHOWN**
FEEDBACK OF PEGAWAI PENILAI	
Signature of Pegawai Penilai Name	DATE